

PROBABILITY AND STATISTICS

Lecture 5

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Outline I

1 Central Limit Theorem (CLT)

2 CLT for Sample Mean

3 Problems

1 Central Limit Theorem (CLT)

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3 Problems

Galton board

<https://www.youtube.com/shorts/MnBBV73KbDo>

Central Limit Theorem - Explained (4 min 40 sec)

<https://www.youtube.com/watch?v=nAfjT-tWWzc>

Real-world application of the Central Limit Theorem (CLT)(7 min 26 sec)

<https://www.youtube.com/watch?v=N7wW1dlmMaE>

CLT for Sum of IID Random Variables

Theorem: Central Limit Theorem for IID Sums

Let $X_1, X_2, \dots, X_n$ be a sequence of **independent and identically distributed (i.i.d.)** random variables with finite mean $\mu = \mathbb{E}[X_i]$ and finite variance $\sigma^2 = \text{Var}(X_i) > 0$.

Let $S_n = \sum_{i=1}^n X_i$. As $n \rightarrow \infty$, the cdf of the normalized sum (Z-score) converges to the standard normal distribution:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z \right) = \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

Key Insight: The distribution of the sum approaches normality regardless of the underlying shape of the individual X_i distributions (whether skewed, uniform, or discrete), provided n is sufficiently large ($n \geq 30$ as a general rule of thumb).

CLT for Sum of IID Random Variables

- **Expectation of the Sum:**

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = n\mu$$

- **Variance of the Sum (due to independence):**

$$\text{Var}(S_n) = \text{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2 \implies \sqrt{\text{Var}(S_n)} = \sigma\sqrt{n}$$

- **Standardization Process:**

$$Z_n = \frac{S_n - \mathbb{E}[S_n]}{\sqrt{\text{Var}(S_n)}} = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Example of Application CLT

Example

An elevator has a maximum weight capacity of 4,000 kg. The weights of individual passengers are i.i.d. random variables with a mean of $\mu = 85$ kg and a standard deviation of $\sigma = 15$ kg.

If 45 passengers board the elevator, what is the probability that their combined weight exceeds the safe operating limit?

1. Identify parameters and distributions

- Number of components: $n = 45$ (Since $45 \geq 30$, CLT is highly applicable).
- Individual parameters: $\mu = 85$, $\sigma = 15$.
- Total weight: $S_{45} = \sum_{i=1}^{45} X_i$.

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Example

2. Mean and Variance of the Total Sum S_{45}

$$\mathbb{E}[S_{45}] = n\mu = 45 \times 85 = 3825 \text{ kg}$$

$$\text{Var}(S_{45}) = n\sigma^2 = 45 \times 15^2 = 10125$$

$$\sqrt{\text{Var}(S_{45})} = \sqrt{10125} \approx 100.62 \text{ kg}$$

3. Standardize and evaluate the probability

$$\begin{aligned}\mathbb{P}(S_{45} > 4,000) &= \mathbb{P}\left(\frac{S_{45} - n\mu}{\sigma\sqrt{n}} > \frac{4000 - 3825}{100.62}\right) \\ &\approx \mathbb{P}(Z > 1.74) \\ &= 1 - \Phi(1.74) \\ &= 1 - 0.9591 = 0.0409\end{aligned}$$

Conclusion: There is approximately a **4.09% chance** that the elevator will be overloaded.

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<https://www.youtube.com/watch?v=3lu7H6EwFSU>

Types of Statistics & Core Concepts

Frequentist Statistics

Bases conclusions on the **long-run frequency** of repeatable events. Assumes parameters are fixed unknowns.

Bayesian Statistics

Updates **prior beliefs** using newly observed data. Assumes parameters are random variables.

Feature	Frequentist	Bayesian
Probability	Limiting frequency	Degree of belief
Parameters	Fixed unknowns	Random variables
Inference	Hypothesis tests, CIs	Posterior distribution

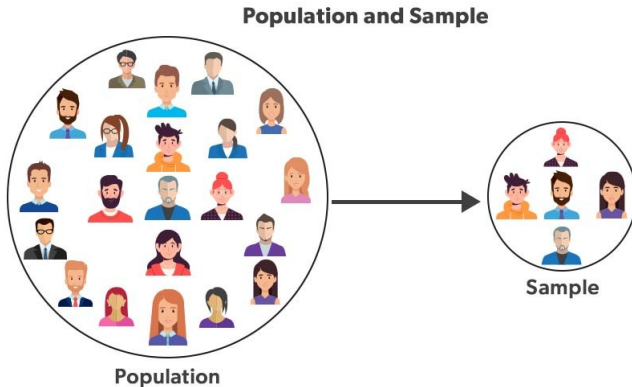
Statistical Approaches:

- **Descriptive:** Summarizes sample data (sample mean, sample variance, sample covariance and so on.)
- **Inferential:** Draws conclusions about the population.

Population vs. Sample:

Population vs. Sample:

- **Population:** The entire group of interest (usually people, objects, transactions, or events that we are interested in studying)
- **Sample:** A subset used to make inferences.



CLT for Sample Mean

Theorem: CLT for Sample Means

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with expected value μ and finite variance σ^2 . The sample mean is defined as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

As $n \rightarrow \infty$, the sampling distribution of $\bar{X}_n$ converges asymptotically to a Normal distribution:

$$\bar{X}_n \xrightarrow{d} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \implies Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Explanation

- **Expectation of sample mean:**

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{\sum_{i=1}^n \mathbb{E}[X_i]}{n} = \frac{n\mu}{n} = \mu$$

- **Variance of the sample mean:**

$$\text{Var}(S_n) = \text{Var}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

- **St.deviation of the sample mean:**

$$\sqrt{\text{Var}(\bar{X}_n)} = \frac{\sigma}{\sqrt{n}}.$$

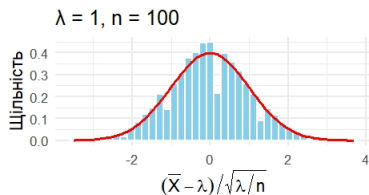
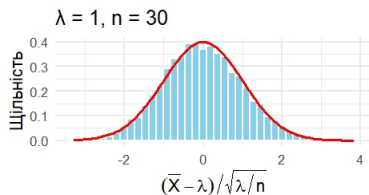
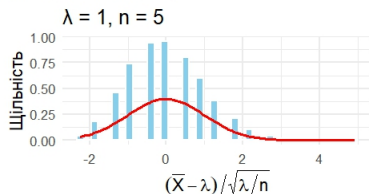
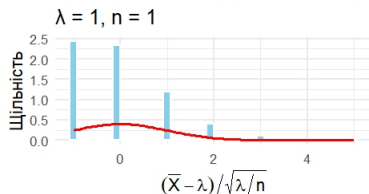
- **Standardization Process:**

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Simulation of CLT

Number of iterations 10000

Центральна гранична теорема для Poisson($\lambda=1$)



Example

Example

The lightbulbs produced by a factory have a mean lifetime of $\mu = 800$ hours and a standard deviation of $\sigma = 40$ hours. The distribution of lifetimes is highly left-skewed.

A quality control inspector selects a random sample of $n = 64$ lightbulbs. What is the probability that the sample mean lifetime $\bar{X}$ is less than 790 hours?

- Despite the distribution being left-skewed, $n = 64 \geq 30$, ensuring the sample mean $\bar{X}$ is normally distributed per the CLT.
- Standard Error (SE) calculation:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{64}} = \frac{40}{8} = 5 \text{ hours}$$

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Solution

We want to compute $\mathbb{P}(\bar{X} < 790)$.

Standardize the sample mean variable:

$$\begin{aligned}\mathbb{P}(\bar{X} < 790) &= \mathbb{P}\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{790 - 800}{5}\right) \\ &= \mathbb{P}\left(Z < \frac{-10}{5}\right) \\ &= \mathbb{P}(Z < -2.00)\end{aligned}$$

Find the standard normal table value:

$$\mathbb{P}(Z < -2.00) = \Phi(-2.00) = 0.0228$$

Conclusion: There is a **2.28% probability** that the sample average lifetime for the 64 inspected lightbulbs will fall below 790 hours.

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Problems

Problem 7

A gambler flips a fair coin n times. How many times must the coin be flipped so that, with a probability of at least 0.95, the sample proportion of "Heads" deviates from 0.5 by no more than 0.01? (Use Chebyshev's Inequality).

Problem 8

A delivery truck transports 100 cargo boxes. The weight of each box is an independent random variable with a mean of $\mu = 20$ kg and a standard deviation of $\sigma = 2$ kg. What is the probability that the total cargo weight exceeds 2,050 kg?

Problem 9

An insurance company underwrites a specialized premium policy for professional digital creators against targeted cyber ransom attacks. The insurance firm has a portfolio of $n = 10,000$ independent creators. For any single creator over a 1-year timeline, the payout distribution is highly skewed: there is a 99.5% chance of zero claims, but a 0.5% chance of a catastrophic breach requiring a maximum payout of \$50,000. To fund these claims, the company charges each creator an annual premium of \$300.

- Calculate the expected value (μ) and standard deviation (σ) of a claim payout for an individual creator.
- Define the distribution of the total aggregate claims (S_n) across all 10,000 policies using the Central Limit Theorem.
- Calculate the probability that the insurance company's total annual claim payouts will exceed the total collected revenue from premiums (causing a net financial loss).

Questions?